Vendor Performance Analysis

- NIKHIL LANKE

Project Vision & Objectives

Visual Transformation

Turn complicated raw data into easy-to-understand and interactive charts or visuals so people can quickly see patterns and insights.

Decision Support

Make data easy for stakeholders to understand so they can make quicker and better business decisions

Easy Analysis

Provide an easy-to-use dashboard that allows people without technical knowledge to understand and use advanced data analysis.

Goals

•Analyze sales and inventory data to evaluate overall business performance

•Identify underperforming brands for possible pricing or promotional improvements

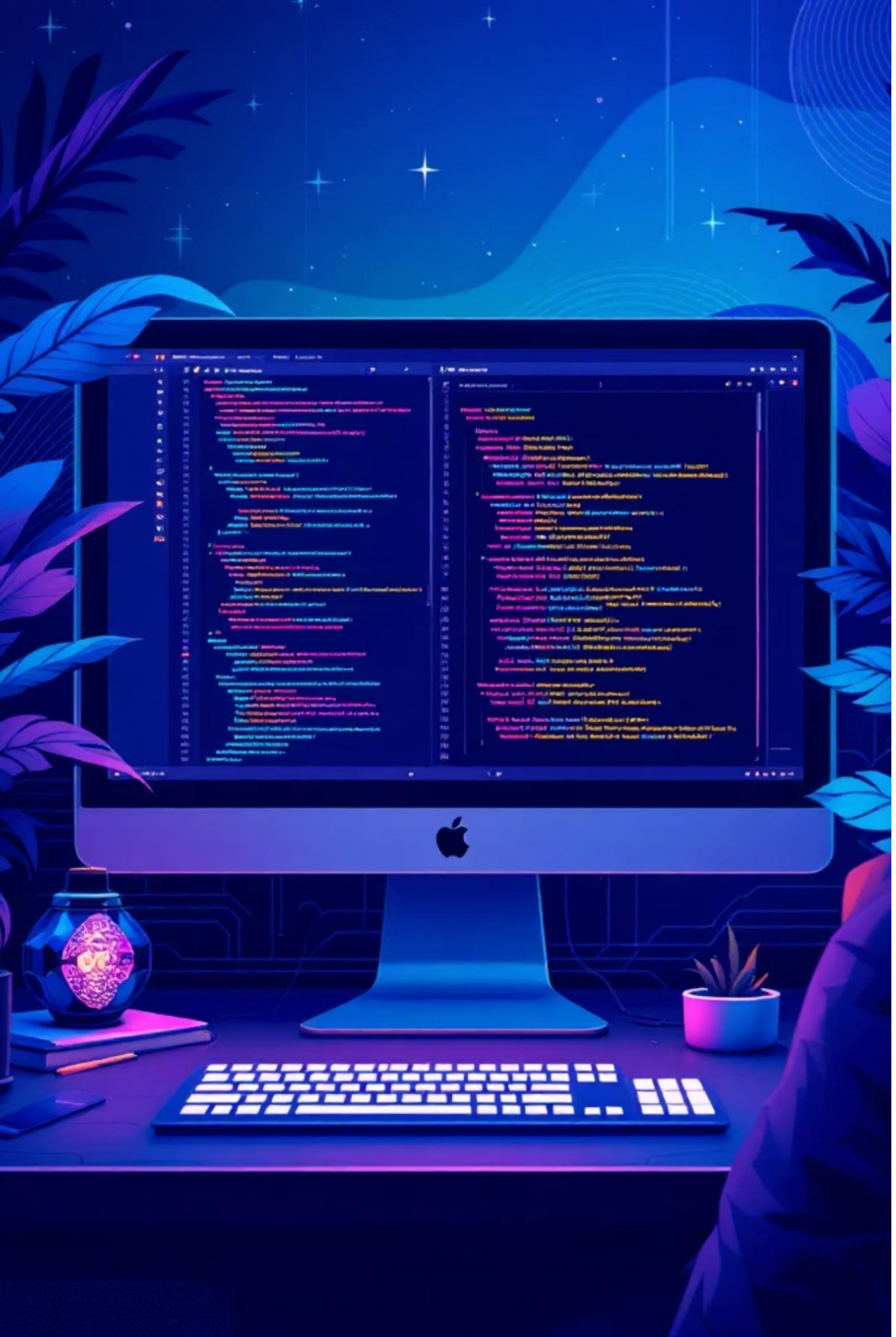
•Determine top vendors contributing to sales and profit

•Study the impact of bulk purchasing on cost and profitability

•Evaluate inventory turnover to improve stock management and reduce holding costs

•Compare profit margins between high-performing and low-performing vendors

•Provide data-driven recommendations to enhance profitability and efficiency



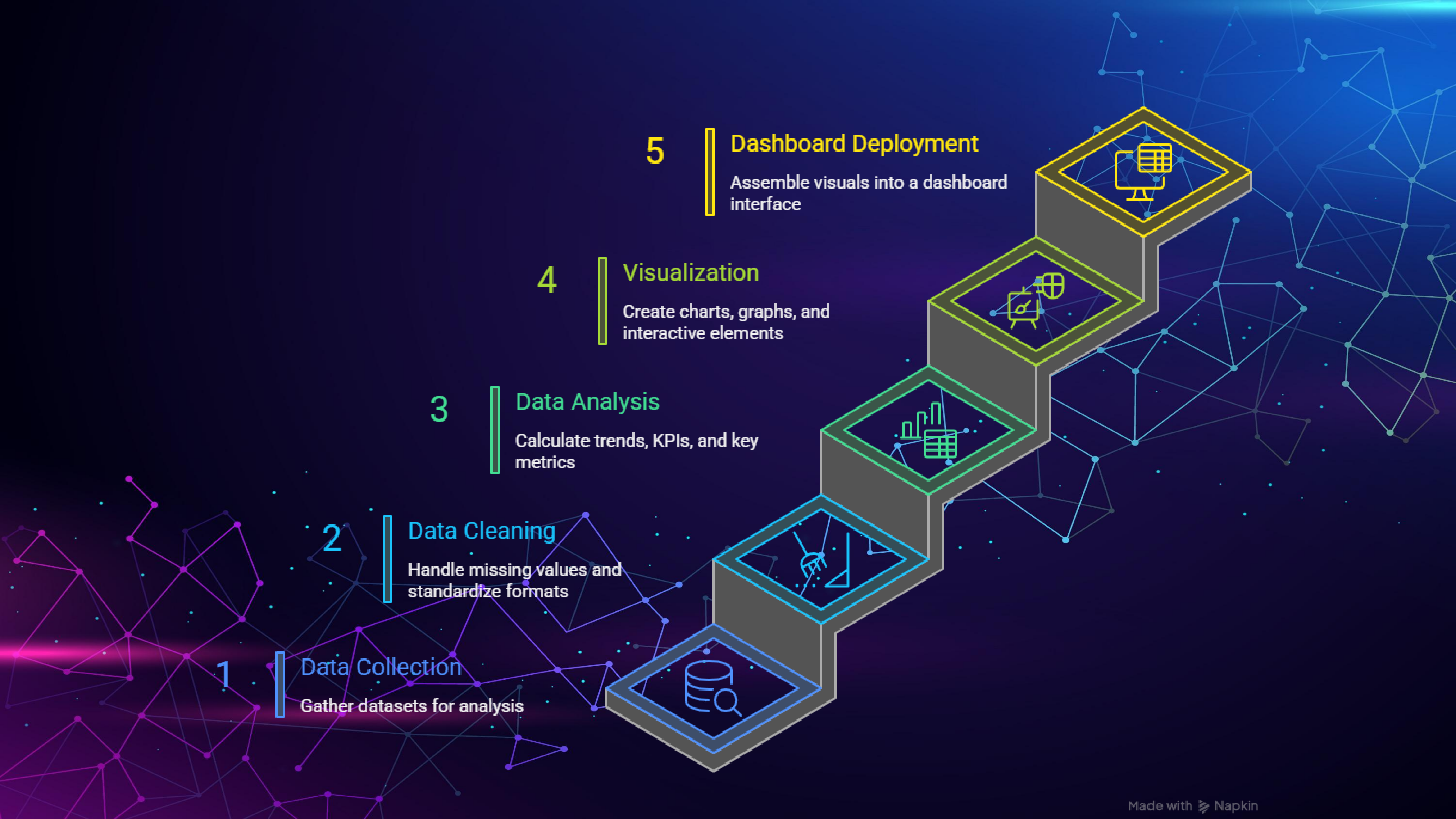
Technology Stack

Core Technologies

- *Python for data manipulation and analysis*
- *Pandas for data transformation*
- *NumPy for numerical computing*
- *Jupyter Notebook for creating Visualizations*
- *SQL for creating tables, Perform some operations on it*

Visualization

- *Tableau*
- *Power BI*


5
Dashboard Deployment
Assemble visuals into a dashboard interface
4
Visualization
Create charts, graphs, and interactive elements
3
Data Analysis
Calculate trends, KPIs, and key metrics
2
Data Cleaning
Handle missing values and standardize formats
1
Data Collection
Gather datasets for analysis
Made with Napkin

Data Collection

Begin Inventory = 206530

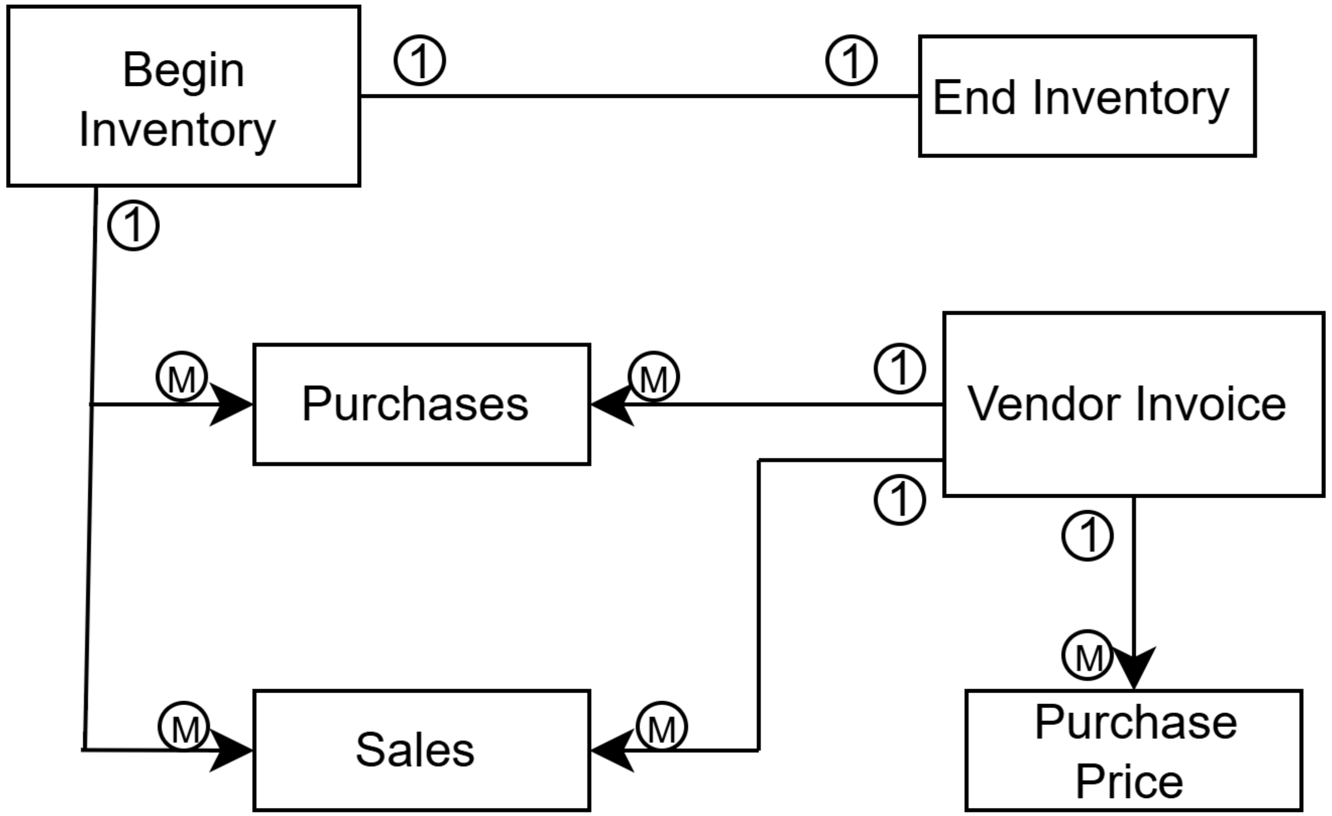
Purchases = 1048576

Sales = 1048576

Vendor Invoice = 5544

Purchase Price = 12262

End Inventory = 224490



Using SQL in Jupyter Notebook

```python
import pandas as pd
import sqlite3
```

```python
# create the database connection
conn = sqlite3.connect('inventory.db')
```

```python
#checking tables present in the database
tables = pd.read_sql_query("SELECT name FROM sqlite_master WHERE type='table'",conn)
tables
```

	name
0	begin_inventory
1	end_inventory
2	purchases
3	purchase_prices
4	sales
5	vendor_invoice

```python
for table in tables['name']:
    print('-'*50, f'{table}','-'*50)
    print('Count of records:', pd.read_sql(f"select count(*) as count from {table}", conn)['count' ].values[0])
```

```
-------------------------------------------------- begin_inventory --------------------------------------------------
Count of records: 206529
-------------------------------------------------- end_inventory --------------------------------------------------
Count of records: 224489
-------------------------------------------------- purchases --------------------------------------------------
Count of records: 2372474
-------------------------------------------------- purchase_prices --------------------------------------------------
Count of records: 12261
-------------------------------------------------- sales --------------------------------------------------
Count of records: 12825363
-------------------------------------------------- vendor_invoice --------------------------------------------------
Count of records: 5543
```

```
vendor_sales_summary = pd.read_sql_query("""WITH
FreightSummary AS (
SELECT
VendorNumber,
SUM(Freight) AS FreightCost
FROM vendor_invoice
GROUP BY VendorNumber
),

PurchaseSummary AS (
SELECT
p.VendorNumber,
p. VendorName,
p.Brand,
p.Description,
p.PurchasePrice,
pp.Price AS ActualPrice,
pp.Volume,
SUM(p.Quantity) AS TotalPurchaseQuantity,
SUM(p.Dollars) AS TotalPurchaseDollars
FROM purchases p
JOIN purchase_prices pp
ON p.Brand = pp.Brand
WHERE p.PurchasePrice > 0
GROUP BY p.VendorNumber, p.VendorName, p.Brand,
p.Description, p.PurchasePrice, pp.Price, pp.Volume
),

SalesSummary AS (
SELECT
VendorNo,
Brand,
SUM(SalesQuantity) AS TotalSalesQuantity,
SUM(SalesDollars) AS TotalSalesDollars,
SUM(SalesPrice) AS TotalSalesPrice,
SUM(ExciseTax) AS TotalExciseTax
FROM sales
GROUP BY VendorNo, Brand
)

SELECT
ps.VendorNumber,
ps.VendorName,
ps.Brand,
ps.Description,
ps.PurchasePrice,
ps.ActualPrice,
ps.Volume,
ps.TotalPurchaseQuantity,
ps.TotalPurchaseDollars,
ss.TotalSalesQuantity,
ss.TotalSalesDollars,
ss.TotalSalesPrice,
Ss.TotalExciseTax,
fs.FreightCost
FROM PurchaseSummary ps
LEFT JOIN SalesSummary ss
ON ps.VendorNumber = ss.VendorNo
AND ps.Brand = ss.Brand
LEFT JOIN FreightSummary fs
ON ps.VendorNumber = fs.VendorNumber
ORDER BY ps.TotalPurchaseDollars DESC""",conn)
```

```
cursor.execute("""CREATE TABLE vendor_sales_summary (
VendorNumber INT,
VendorName VARCHAR(100),
Brand INT,
Description VARCHAR(100),
PurchasePrice DECIMAL(10,2),
ActualPrice DECIMAL(10,2),
Vo lume ,
TotalPurchaseQuantity INT,
TotalPurchaseDollars DECIMAL(15,2),
TotalSalesQuantity INT,
TotalSalesDollars DECIMAL(15,2),
TotalSalesPrice DECIMAL(15,2),
TotalExciseTax DECIMAL(15,2),
FreightCost DECIMAL(15,2),
GrossProfit DECIMAL(15,2),
ProfitMargin DECIMAL(15,2),
StockTurnover DECIMAL(15,2),
SalesToPurchaseRatio DECIMAL(15,2),
PRIMARY KEY (VendorNumber, Brand)
);
""")
```

Final Table (Vendor_Performance_Analysis)

VendorNumber	VendorName	Brand	Description	PurchasePrice	ActualPrice	Vo	TotalPurchaseQuantity	TotalPurchaseDollars	TotalSalesQuantity	TotalSalesDollars	TotalSalesPrice	TotalExciseTax	FreightCost	GrossProfit	ProfitMargin	StockTurnover	SalesToPurchaseRatio

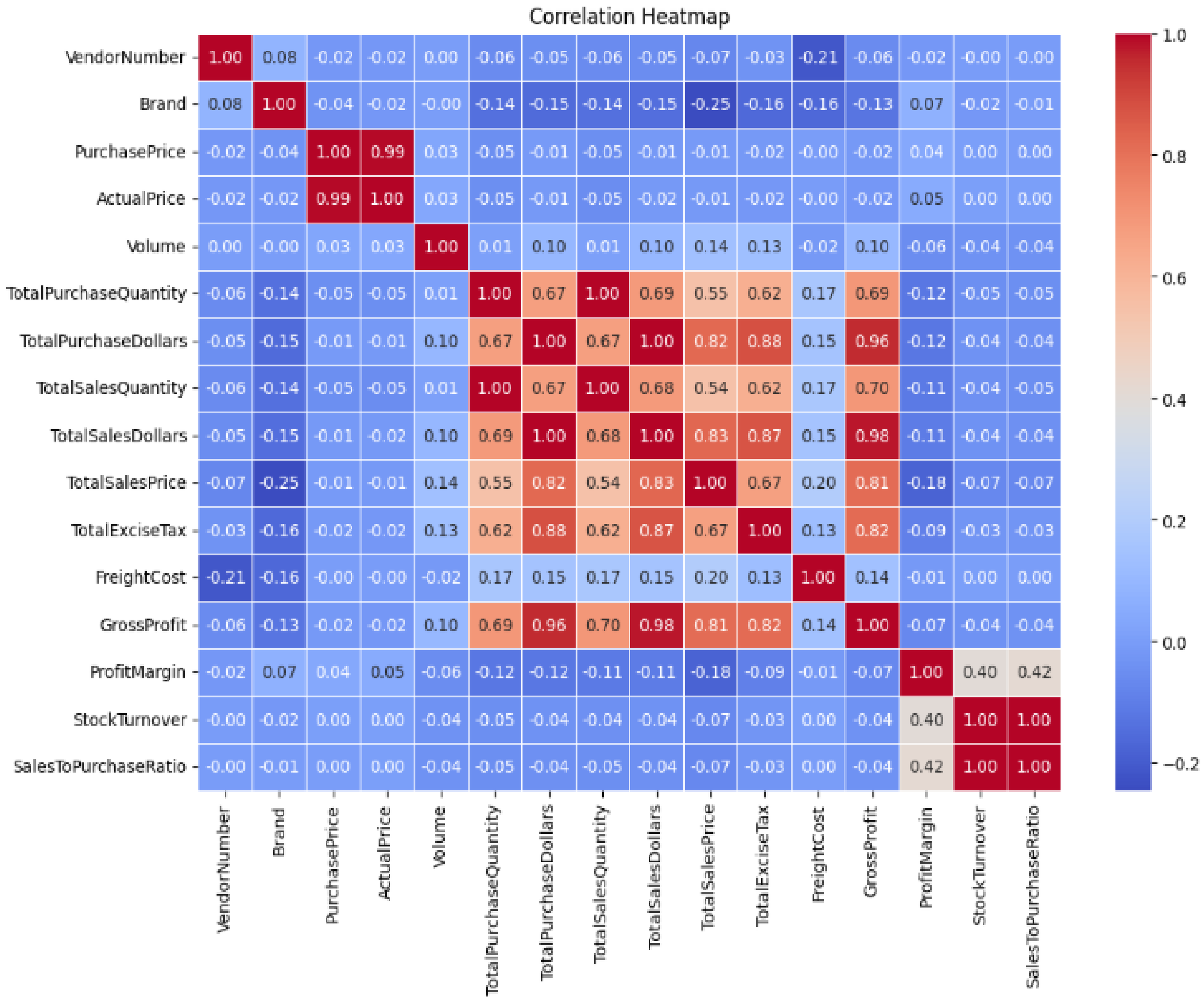
Data Cleaning

To enhance the reliability of the insights, we removed inconsistent data points where:

•Gross Profit ≤ 0 (to exclude transactions leading to losses).

•Profit Margin ≤ 0 (to ensure analysis focuses on profitable transactions).

•Total Sales Quantity = 0 (to eliminate inventory that was never sold).



Purchase Price vs. Total Sales Dollars & Gross Profit: Weak correlation (-0.012 and -0.016), indicating that price variations do not significantly impact sales revenue or profit.

Total Purchase Quantity vs. Total Sales Quantity: Strong correlation (0.999), confirming efficient inventory turnover.

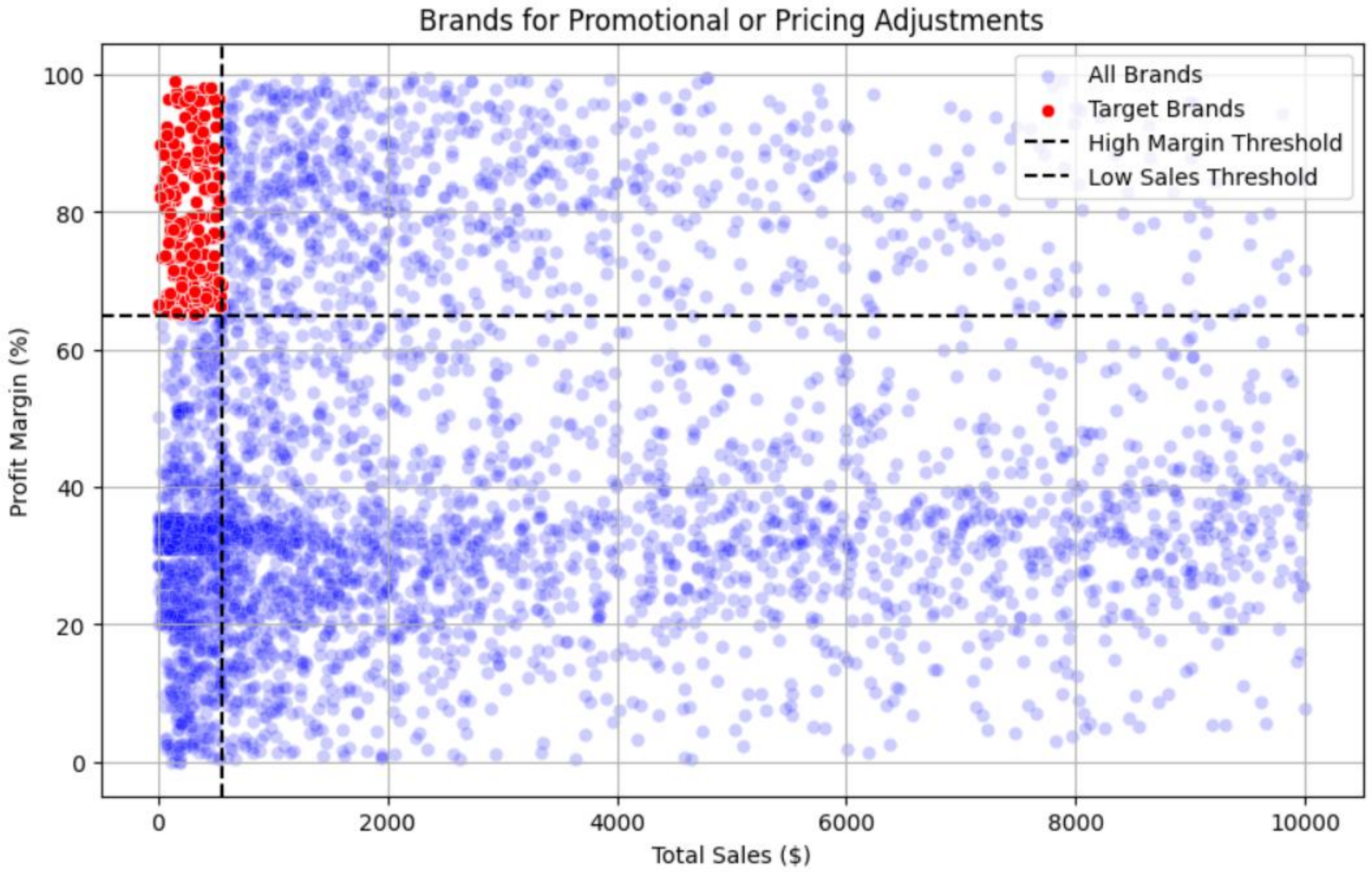
Profit Margin vs. Total Sales Price: Negative correlation (-0.179), suggesting increasing sales prices may lead to reduced margins, possibly due to competitive pricing pressures.

Stock Turnover vs. Gross Profit & Profit Margin: Weak negative correlation (-0.038 & -0.055), indicating that faster stock turnover does not necessarily equate to higher profitability.

Data Analysis

Brands for Promotion or Pricing Strategy

•198 brands have **low sales but high profit margins**
•These brands are **underperforming in volume**
•Opportunity to:
•Increase sales using **promotions**
•Adjust pricing strategically
•Goal: **Boost sales without reducing profit**



Top Vendors Contribution

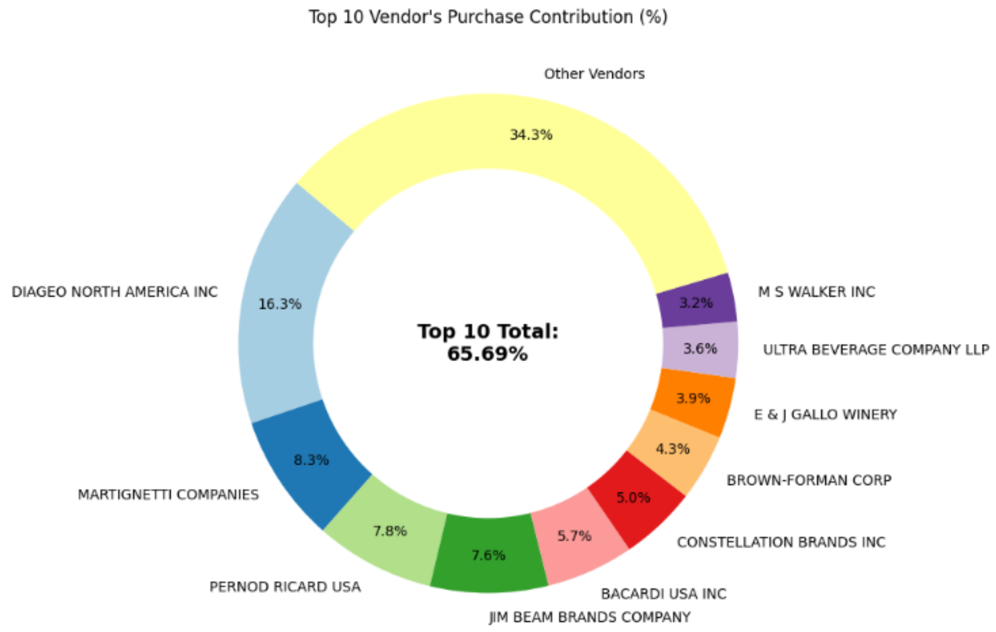
•Top 10 vendors contribute **65.69% of total purchases**
•Remaining vendors contribute only **34.31%**
•Business is **dependent on few vendors**

⚠️ **Risk**

•Supply chain disruption risk
•Limited flexibility

💡 **Solution**

•Diversify vendors
•Build relationships with smaller vendors



Impact of Bulk Purchasing

•Unit cost decreases with larger orders:

- Small: **$39.07**
- Medium: **$15.49**
- Large: **$10.78**

•Bulk orders give **~72% cost reduction**

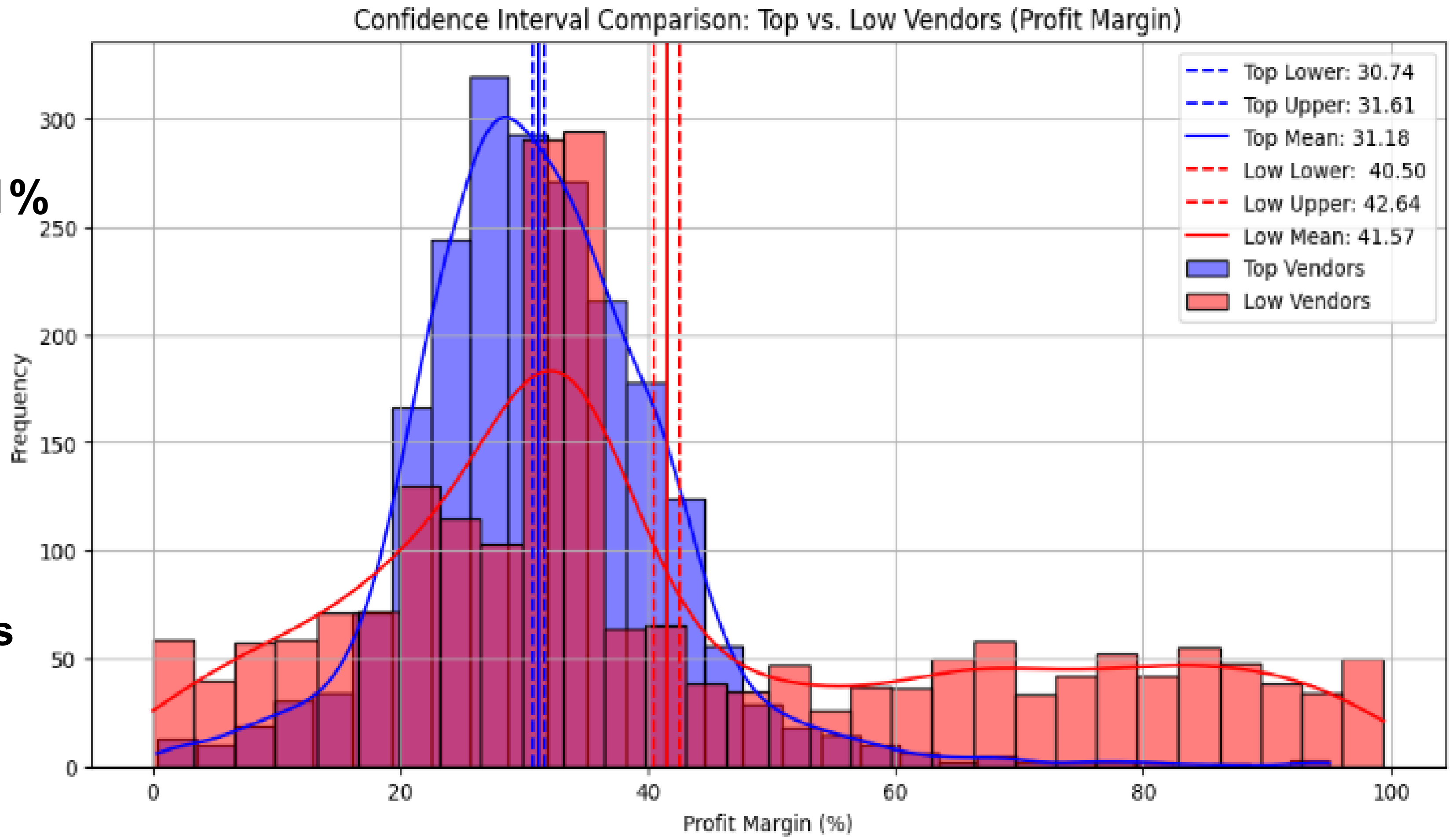
💡 **Insight**

•Encourages **bulk purchasing**
•Increases **profit margins**

	UnitPurchasePrice
OrderSize	
Small	39.068186
Medium	15.486414
Large	10.777625

Profit Margin Analysis: Top vs Low Vendors

- Top vendors have an average profit margin of **~31%**
- Low-performing vendors have a higher margin of **~41%**
- Confidence intervals show:
- Top vendors: **30.74% – 31.61%**
- Low vendors: **40.48% – 42.62%**



- Top vendors focus on **selling more with lower margins**
- Low-performing vendors focus on **higher margins but sell less**

Top vendors make money by selling more, while low vendors try to earn more per item but sell less.

Final Recommendations

•Improve pricing of products that have **low sales but high profit** to increase sales

•Do not depend on only a few vendors, **work with more suppliers**

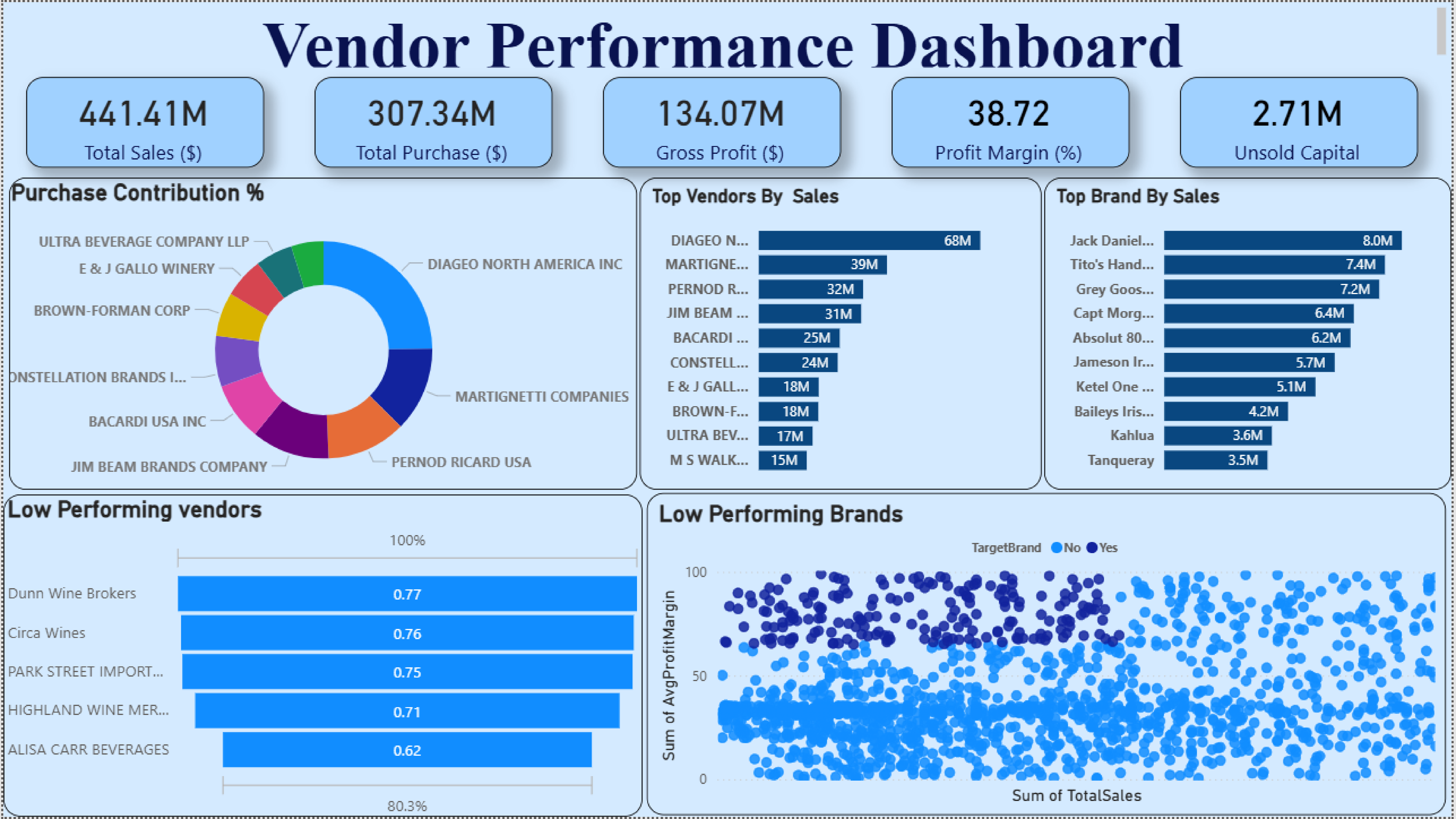
•Buy products in **bulk to reduce cost and increase profit**

•Manage slow-moving products by **offering discounts or reducing stock**

•Improve **marketing and distribution** for low-performing products

•This will help the company **increase profit, reduce risk, and work more efficiently**

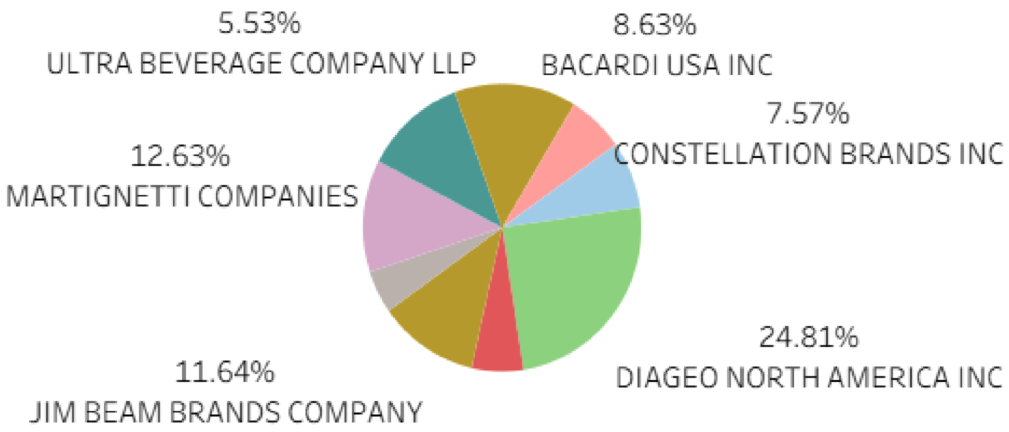
“These recommendations help the company increase sales and profit, reduce risk from depending on few vendors, and manage inventory better. Overall, it will improve business performance.”


Vendor Performance Dashboard
441.41M
Total Sales ($)
307.34M
Total Purchase ($)
134.07M
Gross Profit ($)
38.72
Profit Margin (%)
2.71M
Unsold Capital
Purchase Contribution %
ULTRA BEVERAGE COMPANY LLP
E & J GALLO WINERY
BROWN-FORMAN CORP
ONSTELLATION BRANDS I...
BACARDI USA INC
JIM BEAM BRANDS COMPANY
DIAGEO NORTH AMERICA INC
MARTIGNETTI COMPANIES
PERNOD RICARD USA
Top Vendors By Sales
DIAGEO N... 68M
MARTIGNE... 39M
PERNOD R... 32M
JIM BEAM ... 31M
BACARDI ... 25M
CONSTELL... 24M
E & J GALL... 18M
BROWN-F... 18M
ULTRA BEV... 17M
M S WALK... 15M
Top Brand By Sales
Jack Daniel... 8.0M
Tito's Hand... 7.4M
Grey Goos... 7.2M
Capt Morg... 6.4M
Absolut 80... 6.2M
Jameson Ir... 5.7M
Ketel One ... 5.1M
Baileys Iris... 4.2M
Kahlua 3.6M
Tanqueray 3.5M
Low Performing vendors
100%
Dunn Wine Brokers 0.77
Circa Wines 0.76
PARK STREET IMPORT... 0.75
HIGHLAND WINE MER... 0.71
ALISA CARR BEVERAGES 0.62
80.3%
Low Performing Brands
TargetBrand No Yes
100
50
0
Sum of AvgProfitMargin
Sum of TotalSales

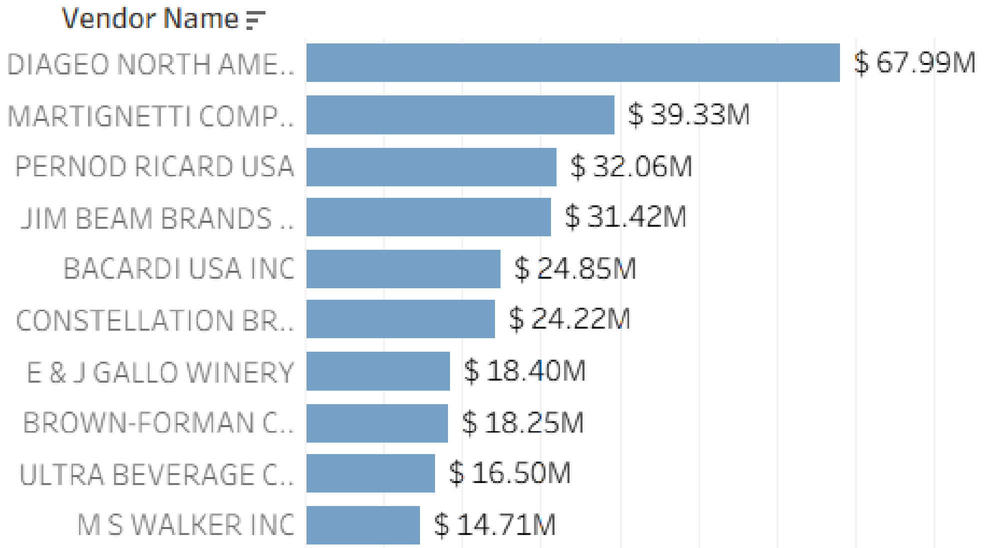
Vendor Performance Dashboard

Total Sales ($)	Total Purchase ($)	Gross Profit ($)	Profit Margin (%)	Unsold Captial
$ 441.41M	$ 307.34M	$134.07M	38.72	$2.71M

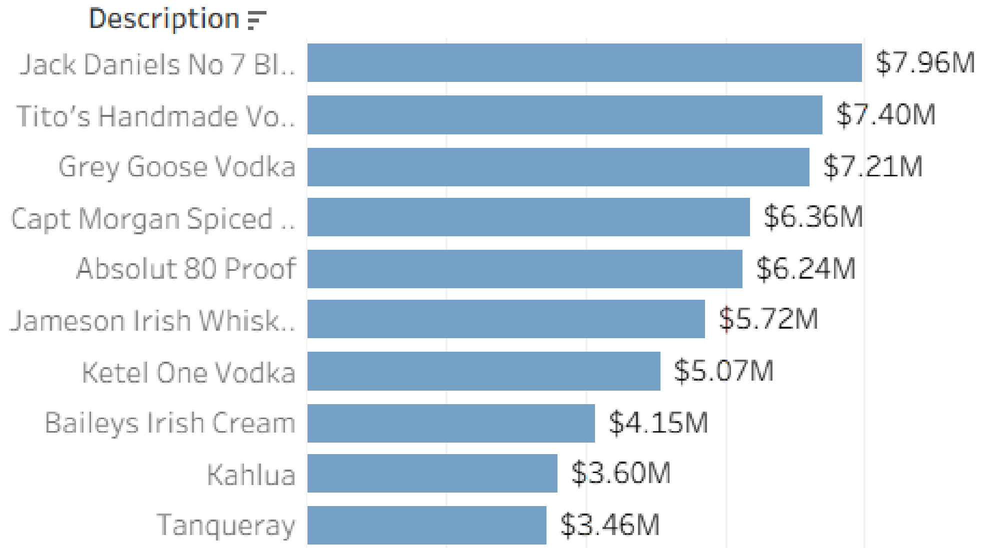
Purchase Contribution %



Top Vendors By Sales

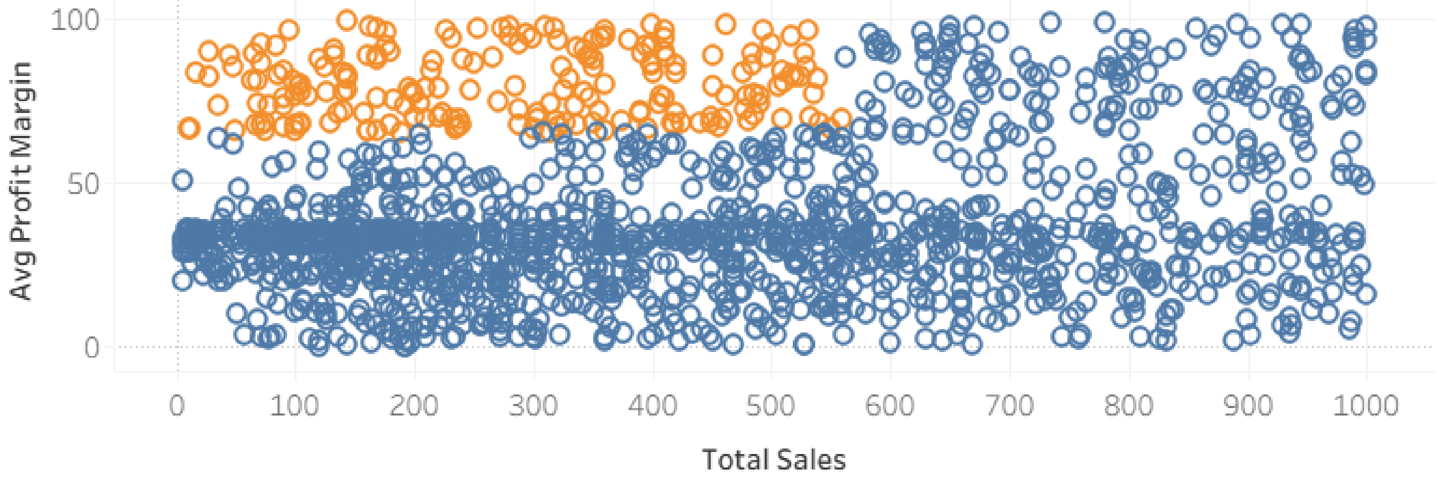


Top Brand By Sales



Low Performing vendors

Low Performing brands



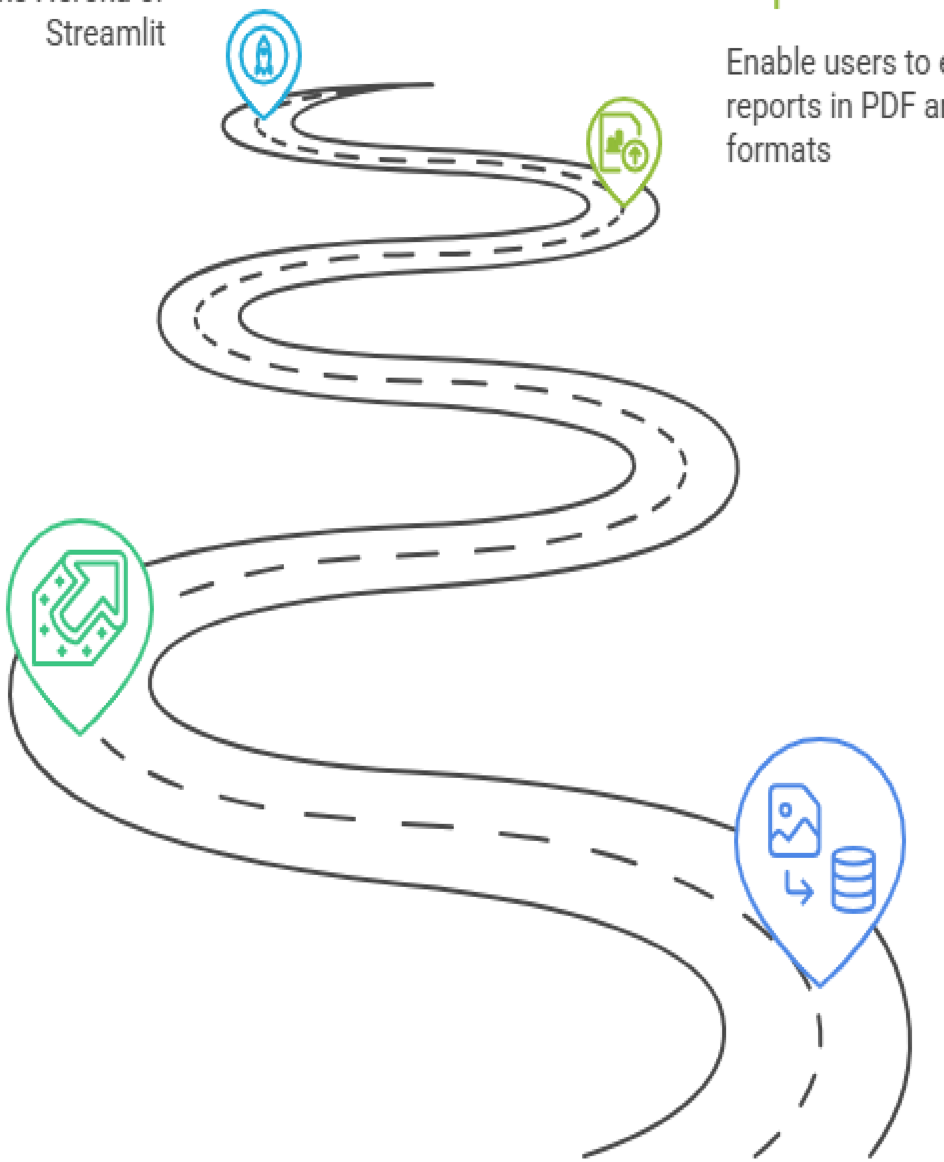
Future Enhancement Sequence

Deploy as Live Web App

Launch the application on platforms like Heroku or Streamlit

Support Exporting Reports

Enable users to export reports in PDF and Excel formats



Enhance UI/UX

Improve the user interface and experience

Add Real-Time Data Updates

Integrate live data feeds into the application